

Deepak Mallampati

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Kent, OH

PROFESSIONAL SUMMARY

Applied AI engineer who ships production agents and has felt the reliability cliff - Agentic Healthcare Claims pipeline pinned multi-agent hallucinations with schema-validated JSON contracts; Progress RAG and grounded eval at $\geq 90\%$ alignment ($\sim 35\%$ retrieval precision, $> 30\%$ hallucination reduction); FastAPI/Docker production discipline ($\sim 30\%$ post-deploy defect reduction). HIPAA-conscious audit + lineage discipline transfers cleanly to risk and compliance agentic AI delivery. Master's in Computer Science (Kent State). Three solo-shipped artifacts, including a Chrome Web Store extension integrating Claude Sonnet in production.

CORE COMPETENCIES

Agentic Workflows • Evaluation and Guardrails • Grounding and RAG • Multi-Agent Systems • Audit-Trail Governance • Python/FastAPI • Customer-Facing Delivery • LLM Applications • Schema Contracts • MLOps

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved retrieval precision $\sim 35\%$, reduced irrelevant retrieval $> 30\%$, grounded-response alignment $> 90\%$.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability $\sim 25\%$, hallucinations cut $> 30\%$.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall $+15-20\%$ on high-risk categories, precision held $> 90\%$ via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects $\sim 30\%$.
- EHR preprocessing (Pandas, NumPy); dataset reliability $> 98\%$, downstream instability -40% .
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP; query time -20% , retrieval latency -25% .
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors -30% , release cycle -35 to -40% .
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence -25% , root-cause resolution -30% .
- Tuned index maintenance and partition-aware queries; reconciliation runtime -15% , deadlocks -15% .
- Supported RBAC and audit logging for financial modules in a regulated environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments $\sim 85\%$.
- Implemented **document Q&A with RAG**; hallucinations $\sim 30\%$, contextual accuracy $> 85\%$.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring.
Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers including Claude Sonnet) with per-provider payload handling for failure isolation. 29 sites supported across Manifest V3 service workers.
TypeScript, Manifest V3, Service Workers, Claude / OpenAI / Google / Ollama

Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt transformation, grounded image generation. ~30% contextual alignment gain, ~25-30% first-pass image success.
FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

YOLOv8 Driver Drowsiness Detection

Real-time computer vision for driver drowsiness; sub-50ms inference on edge hardware. Walk-forward validation and class-weighted training for imbalanced safety data.
PyTorch, YOLOv8, OpenCV, ONNX

EDUCATION

Master of Science, Computer Science - Kent State University	Aug 2023 - May 2025
Focus: Applied Machine Learning, NLP, Database Systems.	
Bachelor of Technology, Computer Science - KL University	Jun 2019 - May 2023
Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives).	

CERTIFICATIONS

Deep Learning Specialization - DeepLearning.AI (Coursera)	2024
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SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++
AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LangGraph, LlamaIndex, OpenAI, Anthropic Claude, Hugging Face Transformers, Diffusers, YOLOv8
LLM / GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search
Backend: FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright
Frontend: React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)
Data: PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake
DevOps: Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda)
Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal